

Advanced Functions & Equations

ANSWER KEY



Function composition

- 1 Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.

$$g(2) = 5, \text{ so } f(g(2)) = 3(5) - 2 = 13. \quad f(2) = 4, \text{ so } g(f(2)) = 4^2 + 1 = 17.$$

- 2 Using the same f and g , find the formula for $(f \circ g)(x)$.

$$(f \circ g)(x) = f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 1.$$

Inverse functions

- 3 Find $f^{-1}(x)$ for $f(x) = 3x - 2$.

$$\text{Let } y = 3x - 2. \text{ Solve for } x: x = \frac{y + 2}{3}. \text{ So } f^{-1}(x) = \frac{x + 2}{3}.$$

- 4 Verify your answer above by computing $f(f^{-1}(5))$.

$$f^{-1}(5) = \frac{5 + 2}{3} = \frac{7}{3}, \text{ and } f\left(\frac{7}{3}\right) = 3\left(\frac{7}{3}\right) - 2 = 7 - 2 = 5. \text{ It matches, confirming the inverse.}$$

Rational equations

- 5 Solve $\frac{2x - 1}{x + 3} = \frac{1}{2}$.

$$\text{Cross-multiply: } 2(2x - 1) = x + 3 \Rightarrow 4x - 2 = x + 3 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3} \text{ (and } x \neq -3, \text{ so this is valid).}$$

- 6 Solve $\frac{3}{x - 1} + 2 = 5$.

$$\frac{3}{x - 1} = 3 \Rightarrow x - 1 = 1 \Rightarrow x = 2 \text{ (and } x \neq 1, \text{ so this is valid).}$$

Factoring higher-degree polynomials

- 7 Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.

By the Factor Theorem, $(x - 1)$ is a factor. Dividing gives

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2). \text{ Roots: } x = 1, 3, -2.$$

- 8 Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.

$$(-2)^3 - 2(-2)^2 - 5(-2) + 6 = -8 - 8 + 10 + 6 = 0. \text{ Yes, } x = -2 \text{ is a root.}$$

Exponential and logarithmic equations

9 Solve $3^{x-1} = 81$.

$$81 = 3^4, \text{ so } x - 1 = 4 \Rightarrow x = 5.$$

10 Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous roots.

Combine: $\log_3[(x+2)(x-2)] = 2 \Rightarrow x^2 - 4 = 3^2 = 9 \Rightarrow x^2 = 13 \Rightarrow x = \pm\sqrt{13}$. The domain requires $x - 2 > 0$, i.e. $x > 2$; since $\sqrt{13} \approx 3.61 > 2$ but $-\sqrt{13} < 2$, only $x = \sqrt{13}$ is valid.

Multiple choice — easy

11 Let $f(x) = x - 9$ and $g(x) = -3x^2 + 7$. Find $f(g(0))$.

a. 0

b. 7

c. -4

d. -2 ✓

$$g(0) = -3(0)^2 + 7 = 7. f(7) = (7) - 9 = -2.$$

12 Let $f(x) = -3x + 7$ and $g(x) = -2x^2 + 5$. Find $f(g(3))$.

a. 46 ✓

b. 48

c. 44

d. -13

$$g(3) = -2(3)^2 + 5 = -13. f(-13) = -3(-13) + 7 = 46.$$

13 Let $f(x) = -2x - 6$ and $g(x) = -x^2 - 1$. Find $f(g(-4))$.

a. 28 ✓

b. -17

c. 26

d. 30

$$g(-4) = -(-4)^2 - 1 = -17. f(-17) = -2(-17) - 6 = 28.$$

14 Let $f(x) = -x + 9$ and $g(x) = 3x^2 + 1$. Find $f(g(-1))$.

a. 3

b. 7

c. 4

d. 5 ✓

$$g(-1) = 3(-1)^2 + 1 = 4. f(4) = -(4) + 9 = 5.$$

15 Let $f(x) = x + 7$ and $g(x) = x^2 - 8$. Find $f(g(-5))$.

a. 26

b. 24 ✓

c. 17

d. 22

$$g(-5) = (-5)^2 - 8 = 17. f(17) = (17) + 7 = 24.$$

16 Let $f(x) = 4x - 8$ and $g(x) = 3x^2 + 6$. Find $f(g(6))$.

a. 448 ✓

b. 450

c. 114

d. 446

$$g(6) = 3(6)^2 + 6 = 114. f(114) = 4(114) - 8 = 448.$$

17 Let $f(x) = 2x + 8$ and $g(x) = -3x^2 - 1$. Find $f(g(2))$.

a. -20

b. -13

c. -18 ✓

d. -16

$$g(2) = -3(2)^2 - 1 = -13. f(-13) = 2(-13) + 8 = -18.$$

18 Let $f(x) = 4x + 4$ and $g(x) = -3x^2 - 7$. Find $f(g(-4))$.

a. -218

b. -55

c. -214

d. -216 ✓

$$g(-4) = -3(-4)^2 - 7 = -55. f(-55) = 4(-55) + 4 = -216.$$

19 Let $f(x) = 3x - 1$ and $g(x) = 4x^2 + 1$. Find $f(g(-2))$.

a. 52

b. 50 ✓

c. 17

d. 48

$$g(-2) = 4(-2)^2 + 1 = 17. f(17) = 3(17) - 1 = 50.$$

20 Let $f(x) = -3x + 3$ and $g(x) = -3x^2 + 3$. Find $f(g(1))$.

a. 1

b. 0

c. 3 ✓

d. 5

$$g(1) = -3(1)^2 + 3 = 0. f(0) = -3(0) + 3 = 3.$$

21 Let $f(x) = 4x - 3$ and $g(x) = 2x^2 - 9$. Find $f(g(1))$.

a. -7

b. -33

c. -29

d. -31 ✓

$$g(1) = 2(1)^2 - 9 = -7. f(-7) = 4(-7) - 3 = -31.$$

22 Let $f(x) = -4x - 9$ and $g(x) = -4x^2 - 3$. Find $f(g(-3))$.

a. -39

b. 145

c. 147 ✓

d. 149

$$g(-3) = -4(-3)^2 - 3 = -39. f(-39) = -4(-39) - 9 = 147.$$

23 Let $f(x) = 2x - 1$ and $g(x) = 4x^2 + 2$. Find $f(g(1))$.

a. 13

b. 9

c. 11 ✓

d. 6

$$g(1) = 4(1)^2 + 2 = 6. f(6) = 2(6) - 1 = 11.$$

24 Let $f(x) = -4x - 5$ and $g(x) = 3x^2 - 3$. Find $f(g(0))$.

a. 7 ✓

b. -3

c. 9

d. 5

$$g(0) = 3(0)^2 - 3 = -3. f(-3) = -4(-3) - 5 = 7.$$

25 Let $f(x) = 2x + 3$ and $g(x) = 4x^2 - 8$. Find $f(g(5))$.

a. 187 ✓

b. 185

c. 189

d. 92

$$g(5) = 4(5)^2 - 8 = 92. f(92) = 2(92) + 3 = 187.$$

Multiple choice — medium

26 Solve $\frac{4x - 3}{x - 8} = 5$.

a. 38

b. 36

c. 37 ✓

d. -37

Cross-multiply: $4x - 3 = 5(x - 8)$, so $-x = -37$, giving $x = 37$ (and $x \neq 8$, so this is valid).

27 Solve $\frac{2x-4}{x+4} = -2$.

a. -1 ✓

b. 0

c. 1

d. -2

Cross-multiply: $2x - 4 = -2(x + 4)$, so $4x = -4$, giving $x = -1$ (and $x \neq -4$, so this is valid).

28 Solve $\frac{4x+5}{x-8} = -1$.

a. $-\frac{2}{5}$

b. $\frac{8}{5}$

c. $\frac{3}{5}$ ✓

d. $-\frac{3}{5}$

Cross-multiply: $4x + 5 = -1(x - 8)$, so $5x = 3$, giving $x = \frac{3}{5}$ (and $x \neq 8$, so this is valid).

29 Solve $\frac{3x+4}{x-9} = -2$.

a. $-\frac{14}{5}$

b. $\frac{14}{5}$ ✓

c. $\frac{9}{5}$

d. $\frac{19}{5}$

Cross-multiply: $3x + 4 = -2(x - 9)$, so $5x = 14$, giving $x = \frac{14}{5}$ (and $x \neq 9$, so this is valid).

30 Solve $\frac{3x-5}{x+6} = 1$.

a. $-\frac{11}{2}$

b. $\frac{13}{2}$

c. $\frac{11}{2}$ ✓

d. $\frac{9}{2}$

Cross-multiply: $3x - 5 = 1(x + 6)$, so $2x = 11$, giving $x = \frac{11}{2}$ (and $x \neq -6$, so this is valid).

31 Solve $\frac{3x - 6}{x + 1} = -4$.

a. $\frac{9}{7}$

b. $\frac{2}{7}$ ✓

c. $-\frac{5}{7}$

d. $-\frac{2}{7}$

Cross-multiply: $3x - 6 = -4(x + 1)$, so $7x = 2$, giving $x = \frac{2}{7}$ (and $x \neq -1$, so this is valid).

32 Solve $\frac{x + 9}{x - 2} = -4$.

a. $-\frac{1}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $-\frac{6}{5}$

Cross-multiply: $x + 9 = -4(x - 2)$, so $5x = -1$, giving $x = -\frac{1}{5}$ (and $x \neq 2$, so this is valid).

33 Solve $\frac{-3x - 7}{x + 3} = 3$.

a. $\frac{8}{3}$

b. $-\frac{11}{3}$

c. $-\frac{5}{3}$

d. $-\frac{8}{3}$ ✓

Cross-multiply: $-3x - 7 = 3(x + 3)$, so $-6x = 16$, giving $x = -\frac{8}{3}$ (and $x \neq -3$, so this is valid).

34 Solve $\frac{-4x + 5}{x - 2} = -3$.

- a. 0
- b. 1
- c. -2
- d. -1 ✓

Cross-multiply: $-4x + 5 = -3(x - 2)$, so $-x = 1$, giving $x = -1$ (and $x \neq 2$, so this is valid).

35 Solve $\frac{-2x + 5}{x - 9} = 1$.

- a. $\frac{17}{3}$
- b. $-\frac{14}{3}$
- c. $\frac{11}{3}$
- d. $\frac{14}{3}$ ✓

Cross-multiply: $-2x + 5 = 1(x - 9)$, so $-3x = -14$, giving $x = \frac{14}{3}$ (and $x \neq 9$, so this is valid).

36 Solve $\frac{-x + 2}{x - 3} = 3$.

- a. $\frac{7}{4}$
- b. $\frac{11}{4}$ ✓
- c. $-\frac{11}{4}$
- d. $\frac{15}{4}$

Cross-multiply: $-x + 2 = 3(x - 3)$, so $-4x = -11$, giving $x = \frac{11}{4}$ (and $x \neq 3$, so this is valid).

37 Solve $\frac{3x + 3}{x - 7} = -2$.

- a. $\frac{11}{5}$ ✓
- b. $\frac{6}{5}$
- c. $-\frac{11}{5}$
- d. $\frac{16}{5}$

Cross-multiply: $3x + 3 = -2(x - 7)$, so $5x = 11$, giving $x = \frac{11}{5}$ (and $x \neq 7$, so this is valid).

38 Solve $\frac{4x+6}{x-7} = -4$.

a. $\frac{11}{4}$ ✓

b. $-\frac{11}{4}$

c. $\frac{15}{4}$

d. $\frac{7}{4}$

Cross-multiply: $4x + 6 = -4(x - 7)$, so $8x = 22$, giving $x = \frac{11}{4}$ (and $x \neq 7$, so this is valid).

39 Solve $\frac{-x+9}{x-1} = 2$.

a. $\frac{11}{3}$ ✓

b. $\frac{14}{3}$

c. $\frac{8}{3}$

d. $-\frac{11}{3}$

Cross-multiply: $-x + 9 = 2(x - 1)$, so $-3x = -11$, giving $x = \frac{11}{3}$ (and $x \neq 1$, so this is valid).

40 Solve $\frac{-3x+2}{x-6} = -2$.

a. -10 ✓

b. -9

c. 10

d. -11

Cross-multiply: $-3x + 2 = -2(x - 6)$, so $-x = 10$, giving $x = -10$ (and $x \neq 6$, so this is valid).

Multiple choice — hard

41 Find $f^{-1}(x)$ for $f(x) = 2x - 5$.

a. $\frac{x+5}{2}$ ✓

b. $\frac{x+5}{-2}$

c. $2x - 5$

d. $\frac{x-5}{2}$

Let $y = 2x - 5$. Solve for x : $x = \frac{y+5}{2}$. So $f^{-1}(x) = \frac{x+5}{2}$.

42 Find $f^{-1}(x)$ for $f(x) = 3x + 2$.

a. $3x + 2$

b. $\frac{x-2}{3}$ ✓

c. $\frac{x+2}{3}$

d. $\frac{x-2}{-3}$

Let $y = 3x + 2$. Solve for x : $x = \frac{y-2}{3}$. So $f^{-1}(x) = \frac{x-2}{3}$.

43 Find $f^{-1}(x)$ for $f(x) = 4x + 7$.

a. $\frac{x-7}{-4}$

b. $4x + 7$

c. $\frac{x-7}{4}$ ✓

d. $\frac{x+7}{4}$

Let $y = 4x + 7$. Solve for x : $x = \frac{y-7}{4}$. So $f^{-1}(x) = \frac{x-7}{4}$.

44 Find $f^{-1}(x)$ for $f(x) = -4x + 4$.

a. $-4x + 4$

b. $\frac{x+4}{-4}$

c. $\frac{x-4}{-4}$ ✓

d. $\frac{x-4}{4}$

Let $y = -4x + 4$. Solve for x : $x = \frac{y-4}{-4}$. So $f^{-1}(x) = \frac{x-4}{-4}$.

45 Find $f^{-1}(x)$ for $f(x) = 2x - 8$.

a. $\frac{x+8}{2}$ ✓

b. $2x - 8$

c. $\frac{x+8}{-2}$

d. $\frac{x-8}{2}$

Let $y = 2x - 8$. Solve for x : $x = \frac{y+8}{2}$. So $f^{-1}(x) = \frac{x+8}{2}$.

46 Given that $x = 4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$ ✓

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

Dividing out $(x - 4)$ gives $x^2 + 4x + 3 = 0$, which factors as $(x - (-1))(x - (-3)) = 0$, so $x = -1$ or $x = -3$. Answer: $(-3, -1)$.

47 Given that $x = -6$ is a root of $x^3 + 13x^2 + 54x + 72 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -3)$

b. $(-4, -4)$

c. $(4, 3)$

d. $(-4, -3)$ ✓

Dividing out $(x + 6)$ gives $x^2 + 7x + 12 = 0$, which factors as $(x - (-3))(x - (-4)) = 0$, so $x = -3$ or $x = -4$. Answer: $(-4, -3)$.

- 48 Given that $x = 3$ is a root of $x^3 - 10x^2 + 27x - 18 = 0$, use the Factor Theorem to find the other two roots.

- a. (2, 6)
- b. (1, 5)
- c. (-1, -6)
- d. (1, 6) ✓

Dividing out $(x - 3)$ gives $x^2 - 7x + 6 = 0$, which factors as $(x - 1)(x - 6) = 0$, so $x = 1$ or $x = 6$.
Answer: (1, 6).

- 49 Given that $x = -3$ is a root of $x^3 + 2x^2 - 33x - 90 = 0$, use the Factor Theorem to find the other two roots.

- a. (-4, 6)
- b. (5, -6)
- c. (-5, 6) ✓
- d. (-5, 5)

Dividing out $(x + 3)$ gives $x^2 - x - 30 = 0$, which factors as $(x - 6)(x - (-5)) = 0$, so $x = 6$ or $x = -5$.
Answer: (-5, 6).

- 50 Given that $x = 2$ is a root of $x^3 + 6x^2 - 9x - 14 = 0$, use the Factor Theorem to find the other two roots.

- a. (-6, -1)
- b. (-7, -2)
- c. (7, 1)
- d. (-7, -1) ✓

Dividing out $(x - 2)$ gives $x^2 + 8x + 7 = 0$, which factors as $(x - (-7))(x - (-1)) = 0$, so $x = -7$ or $x = -1$. Answer: (-7, -1).

51 Solve $2^{x+6} = 1$ for x .

- a. -4
- b. -7
- c. -5
- d. -6 ✓

Since $1 = 2^0$: $x + 6 = 0$, so $x = -6$.

52 Solve $2^{2x+1} = 2$ for x .

- a. 1
- b. 0 ✓
- c. -1
- d. 2

Since $2 = 2^1$: $2x + 1 = 1$, so $2x = 0$, giving $x = 0$.

53 Solve $5^{2x+7} = 5$ for x .

- a. -4
- b. -2
- c. -1
- d. -3 ✓

Since $5 = 5^1$: $2x + 7 = 1$, so $2x = -6$, giving $x = -3$.

54 Solve $3^{2x+6} = 1$ for x .

- a. -1
- b. -2
- c. -4
- d. -3 ✓

Since $1 = 3^0$: $2x + 6 = 0$, so $2x = -6$, giving $x = -3$.

55 Solve $2^x = 8$ for x .

a. 5

b. 2

c. 3 ✓

d. 4

Since $8 = 2^3$: $x = 3$.